Q1

(a) $\neg p$ (b) $p \wedge \neg q$ (c) $p \to q$ (d) $\neg q \to \neg p$

(e) $p \to q$ (f) $q \wedge \neg p$ (g) $q \to p$

Q2. 0: False 1: True

(a)

p	q	$p \oplus q$	$p \wedge q$	$(p \oplus q) \to (p \wedge q)$
0	0	0	0	1
0	1	1	0	0
1	0	1	0	0
1	1	0	1	1

(b)

p	q	$p \leftrightarrow q$	$\neg p \leftrightarrow q$	$(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$
0	0	1	0	1
0	1	0	1	1
1	0	0	1	1
1	1	1	0	1

Q3. p: Logic is diffcult q: many students like logic

s: mathematics is easy r: logic is diffcult

(a) $q \to \neg s$ (b) $\neg s \to \neg q$

(c) $\neg s \vee p$ (d) $\neg r \vee \neg s$

(e) $\neg q \to (\neg s \vee \neg r)$

Q4.

(1) In correct

$\neg p \wedge (p \to q)) \to \neg q$ Premise

$\equiv \neg(\neg p \wedge (p \to q)) \vee (\neg q)$ Useful Law

$\equiv (p \vee \neg(p \to q)) \vee (\neg q)$ Premise

$\equiv (p \vee \neg(\neg p \vee q)) \vee (\neg q)$ Useful Law

$\equiv (p \vee (p \wedge (\neg q))) \vee (\neg q)$ Premise

$\equiv p \vee (\neg q)$ Absorption Law

p	$\neg q$	$p \vee (\neg q)$
0	0	0
0	1	1
1	0	1
1	1	1

Since $p \vee (\neg q)$ is not a tautology, $\neg p \wedge (p \to q) \to \neg q$ is not a tautology

(2) correct.

p	q	r	$(p \vee q) \to r$	$(p \to r) \wedge (q \to r)$
0	0	0	1	1
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	0	0
1	0	1	1	1
1	1	0	0	0
1	1	1	1	1

, So they are equivalent

(3) In correct

when $y \neq 0$, $y=1$ or $y=2$, Since $xy=1$, thus there is not existing a x that both satisfy $xy=1$ when $y=1$ or $y=2$. If $x=c$, then $y=\frac{1}{c}$, but y can be other values different from $\frac{1}{c}$. It's invalid.

(4) Correct. When $n=\pm1, m=\pm2$ or $n=\pm2, m=\pm1$, the statement is correct

Q5(1) p: "You did not finish your homework"

q: "You cannot answer this question".

to prove $\begin{array}{l} p\to q \\ \neg p \\ \hline \therefore \neg q \end{array}$ is invalid.

is to prove $(\neg p \wedge (p\to q)) \to (\neg q)$ is not a tautology

$(\neg p \wedge (p\to q)) \to (\neg q)$

$\equiv (\neg p \wedge (\neg p \vee q)) \to (\neg q)$ Useful Law

$\equiv (\neg p \wedge \neg p) \vee (\neg p \wedge q) \to (\neg q)$ Distributive Law

$\equiv \neg p \vee (\neg p \vee q) \to (\neg q)$ Idempotent Law

$\equiv (\neg p \vee \neg p) \vee q \to \neg q$ Associative Law

$\equiv \neg p \vee q \to \neg q$ Idempotent Law

$\equiv \neg(\neg p \vee q) \vee \neg q$ Useful Law

$\equiv (p \wedge \neg q) \vee \neg q$ Absorption Law

$\equiv \neg q$

when $\neg q$ is false, $(p\wedge \neg q)\vee \neg q$ is false, so it is not a tautology.

(2) p: "All students in this class has submitted their homework".

q: "All students can get 100 in the final exam".

Conclusion: $\neg p \wedge (p\to q) \to (\neg q)$

$\neg p \wedge (p\to q) \to (\neg q)$

$\equiv (\neg p \wedge (\neg p \vee q)) \to (\neg q)$ Useful Law

$\equiv (\neg p \wedge \neg p) \vee (\neg p \wedge q) \to (\neg q)$ Distributive Law

$\equiv \neg p \vee (\neg p \vee q) \to (\neg q)$ Idempotent Law

$\equiv (\neg p \vee \neg p) \vee q \to \neg q$ Associative Law

$\equiv \neg p \vee q \to \neg q$ Idempotent Law

$\equiv \neg(\neg p \vee q) \vee \neg q$ Useful Law

$\equiv (p \wedge \neg q) \vee \neg q$ Absorption Law

$= \neg q$

Since $\neg p$ is not a tautology, the argument is invalid.

Q6. $(\neg r \vee (p \wedge \neg q)) \rightarrow (r \wedge p \wedge \neg q)$

$\equiv \neg(\neg r \vee (p \wedge \neg q)) \vee (r \wedge p \wedge \neg q)$ Useful Law

$\equiv r \wedge \neg(p \wedge \neg q) \vee (r \wedge p \wedge \neg q)$ De Morgan's Law

$\equiv r \wedge (\neg p \vee q) \vee (r \wedge p \wedge \neg q)$ De Morgan's Law

$\equiv r \wedge ((\neg p \vee q) \vee (p \wedge \neg q))$ Distributive Laws

$\equiv r \wedge ((\neg p \vee q) \vee \neg(\neg p \vee q))$ Double Negation Laws

$\equiv r \wedge T$ Negation Laws

$\equiv r$

$\equiv r \vee s$

Q7.(a) $\neg(p \oplus q)$

$\equiv \neg((p \wedge \neg q) \vee (\neg p \wedge q))$

$\equiv \neg(p \wedge \neg q) \wedge \neg(\neg p \wedge q)$ De Morgan's Law

$\equiv (\neg p \vee q) \wedge (p \vee \neg q)$ De Morgan's Law

$\equiv ((p \vee \neg q) \wedge \neg p) \vee ((p \vee \neg q) \wedge q)$ Distributive Law

$\equiv [(\neg p \wedge p) \vee (\neg p \wedge \neg q)] \vee [(q \wedge \neg q) \vee (q \wedge p)]$ Distributive Law

$\equiv (\neg p \wedge \neg q) \vee (q \wedge p)$ Negation Law

$\equiv p \leftrightarrow q$

(b) $\neg(p\to q)\to\neg q$

$\equiv (p\to q)\vee(\neg q)$ Useful Laws

$\equiv (\neg p\vee q)\vee(\neg q)$ Useful Laws

$\equiv \neg p\vee(q\vee\neg q)$ Associative Laws

$\equiv \neg p\vee T$ Negation Laws

$\equiv T$ Domination Laws

(c) $(p\to q)\to((r\to p)\to(r\to q))$

$\equiv \neg(\neg p\vee q)\vee(\neg(\neg r\vee p)\vee(\neg r\vee q))$ Useful Laws

$\equiv (p\wedge\neg q)\vee((r\wedge\neg p)\vee(\neg r\vee q))$ DeMorgan's Laws

$\equiv (p\wedge\neg q)\vee(((r\wedge\neg p)\vee\neg r)\vee q)$ Associative Laws

$\equiv (p\wedge\neg q)\vee(((\neg r\vee r)\wedge(\neg r\vee\neg p))\vee q)$ Distributive Laws

$\equiv (p\wedge\neg q)\vee(\neg r\vee\neg p\vee q)$ Double Negation Laws. Associative Laws

$\equiv (p\wedge\neg q)\vee\neg r\vee(\neg(p\wedge\neg q))$ Commutative Laws

$\equiv (p\wedge\neg q)\vee\neg(p\wedge\neg q)\vee\neg r$ Negation Laws

$\equiv T\vee\neg r$ Domination Laws

$\equiv T$

Q8 (a) $\exists x\,(C(x) \wedge F(x) \wedge P(x))$

(b) $\forall x\,(C(x) \wedge D(x) \wedge F(x))$

(c) $\exists x\,(C(x) \wedge F(x) \wedge \neg D(x))$

(d) $\neg \forall x\,(C(x) \wedge D(x) \wedge F(x))$

(e) $\exists x \exists y \exists z\,(C(x) \wedge D(y) \wedge F(y))$

Q9 Proof: 1. $p \wedge q$ Premise
Premise (i)
2. p Simplification Laws

3. $p \rightarrow \neg(q \wedge r)$ Premise (ii)

4. $\neg q \vee \neg r$ 2.3

5. q Premise (iii) Simplification Laws

6. $\neg r$ 4.5 Premise (iiii)

7. $s \rightarrow r$ Premise (iiiii)

8. $\neg s$

Q10 (a) $\exists n \in N\,(\neg(n^3+6n+5$ is odd $\Rightarrow n$ is even$)$

$\equiv \exists n \in N\,(\neg\,(\neg(n^3+6n+5$ is odd$) \vee (n$ is even$)))$

$\equiv \exists n \in N\,(n^3+6n+5$ is odd $\wedge\ n$ is odd$)$

(b) If n^3+6n+5 is odd. $\Rightarrow n(n^2+6)+5$ is odd

then $n(n^2+6)$ is even. let $n(n^2+6)=2k$

First we prove that an ~~odd~~ even multiply an ~~odd~~ even gets an ~~odd~~ even.

$(2k+1)(2m+1) = 4mk + 2(k+m)+1.$

$2k \times 2m = 4mk.$

So, if n is an odd. n^2+6 is an odd $n(n^2+6)$ is an odd, contradiction

if n is an even. n^2+6 is an even, $n(n^2+6)$ is an even. yes.

thus the original statement is true.

② For the negation statement.

$\exists n \in N(n^3+6n+5$ is odd $\wedge$ n is odd$)$

n is odd, n^2+6 is odd, $n(n^2+6)$ is odd. $n(n^2+6)+5$ is even.

There is a contradiction with the negation statement.

So the negation is false and the origional statement is true.

Q11. Proof: Suppose $a^2+b^2=2n$

thus, $(a+b)^2=a^2+b^2+2ab = 2(n+ab)$

So we should prove that if $(a+b)^2$ is an even, then $a+b$ is also an even.

let $x=2n+1$, and suppose its square is an even.

then $x^2=4n^2+4n+1=4(n^2+n)+1$, which is an odd,

So an odd number's square can't be an even.

thus if $(a+b)^2$ is an even, then $a+b$ is also an even.

Since $(a+b)^2=2(n+ab)$, which is an even,
$a+b$ is also an even.

Q12. Proof: Suppose $\sqrt[3]{2}$ is rational, thus $\sqrt[3]{2}$ can be expressed as:

$\sqrt[3]{2}=\frac{a}{b}$ (a, b are integers and a and b have no common factors)

$2=\frac{a^3}{b^3}$, $a^3=2b^3 \Rightarrow a^3$ is an even

Since a^3 is an even, a is also an even. let $a=2k$

then $8k^3=2b^3 \Rightarrow b^3=4k^3$, which is an even.

b^3 is an even $\Rightarrow b$ is an even

$\Rightarrow$ a and b have common factors, which is contradictional to our propose. Then $\sqrt[3]{2}$ is irrational.